

IQO Bot

Adaptive Multimodal Laboratory Assistant for Academic Environments

Central School-Server AI Hosting • Distributed Low-Cost Bench Nodes • Dual Delivery

Project Members: Berbache Ayoub • Filalli Yacine Siradj Eddin • Malem Mohamed Mahdi

Academic Supervision: Supervised by Pr. Athmani.S

Software Engineering & Applied Intelligent Systems Module

1. Project Overview & Problem Statement

Higher education laboratories in engineering, physical sciences, and technology face three systemic operational challenges:

1. **High Student-to-Instructor Ratio:** A single instructor typically supervises 30 to 40 students simultaneously across 15 to 20 workbench stations, creating instructional bottlenecks and delayed student feedback during practical lab procedures.
2. **Laboratory Safety & Equipment Hazards:** Introductory students frequently risk personal injury or hardware destruction due to incorrect wiring (short circuits, reverse polarity), improper chemical reagent handling, or exceeding rated power limits.
3. **Prohibitive Technology Adoption Costs:** Existing commercial intelligent laboratory robotics require dedicated high-performance edge computing units (e.g., NVIDIA Jetson Orin at $500+ per bench), making campus-wide lab deployments financially infeasible.

The IQO Bot Solution: An adaptive multimodal assistant designed to provide real-time, non-invasive bench guidance. To overcome cost barriers, the AI intelligence is centralized on the existing university server infrastructure, while each student workbench hosts an ultra-low-cost client node ($63 BOM) with stationary non-locomotive kinematics.

2. Core Architectural Paradigm: Central University Server Hosting

Rather than distributing costly edge GPUs across individual student desks, IQO Bot decouples perception and actuation from neural reasoning:

- **Central School Server (Inference Core):** Hosted on existing campus server or workstation hardware. Utilizes open-source quantized 4-bit Small Language Models (e.g., LLaMA-3 8B, Qwen-2.5 7B) served via Ollama or vLLM over the university local area network (LAN). This completely eliminates per-token cloud API fees ($0 ongoing operating cost) and ensures 100% student privacy and zero data leakage.
- **Lightweight Workbench Client Nodes:** Compact, affordable microcomputing units (Raspberry Pi 4 / ESP32-S3) interfacing with a standard USB webcam, microphone array, and 3-DOF articulated pointer arm.
- **Deterministic Safety Guardrails:** Critical laboratory safety rules (e.g., electrical overcurrent thresholds, banned chemical mixtures) are enforced via local deterministic regex/rule filters prior to LLM generation, ensuring zero hallucination on safety-critical directives.

3. Graph 1: System Interaction Lifecycle & State Machine

The continuous operational cycle of IQO Bot follows a five-stage reactive loop from student bench arrival to task completion:

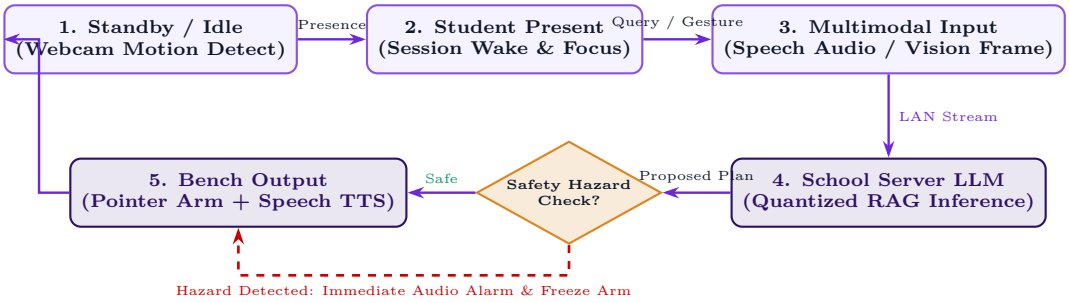


Figure 1: Closed-loop bench interaction pipeline featuring local safety override and centralized server inference.

Technical Specifications & Architecture Master Tables

Comprehensive Hardware Bill of Materials, System Subsystems, and Domain Demonstration Matrix

3.1. Subsystem Architecture & Technology Stack

The complete system is partitioned into modular, loosely coupled subsystems operating across the local area network:

Subsystem	Core Functionality	Software Stack / Libraries	Hardware Allocation	Latency
Workbench Client Node	Captures AV streams, communicates with server via WebSockets, drives servos.	Python 3.10, OpenCV, PyAudio, WebSockets, pigpio.	Raspberry Pi 4 (2GB) or ESP32-S3 bench unit.	< 15 ms
Central AI Server	Hosts quantized open-source LLM, performs RAG retrieval, runs professor admin panel.	Ollama / vLLM, LLaMA-3 8B (Q4_K_M), ChromaDB, FastAPI.	Existing University Server (CPU or single GPU).	180–250 ms
Perception Engine	Person detection, student hands tracking, breadboard/beaker visual inspection.	YOLOv8-nano (CPU optimized), MediaPipe Hands, OpenCV.	Edge client CPU (lightweight) or Server batch.	< 40 ms
Speech Interaction	Keyword wake, local speech-to-text transcription, natural voice synthesis.	Whisper.cpp (quantized small), Piper TTS / espeak-ng.	Shared: Client capture $\rightarrow$ Server transcription.	< 280 ms
Actuator Controller	Computes inverse kinematics for 3-DOF pointer arm and 2-DOF expressive pan-tilt head.	Micro-ROS / Servo PWM controller, Python Kinematics.	3x SG90 / MG90S metal gear servos, I2C driver.	< 30 ms
3D Web Simulator	Digital Twin for remote dry-run simulation and distance learning students.	Three.js, WebGL, WebSockets, HTML5 Canvas.	Student web browser (zero local install).	60 FPS

Table 1: Comprehensive breakdown of IQO Bot software, hardware, and operational latency specifications.

3.2. Hardware Bill of Materials (BOM) & Economic Cost Optimization

Table 2 details the concrete cost optimization achieved by replacing edge-heavy GPU processors with a centralized school-server paradigm:

Functional Component	Proposed Low-Cost Model	Industrial Baseline (Jetson)	Savings / Unit
Edge Compute Processor	Raspberry Pi 4 (2GB RAM) / ESP32-S3 ($35.00)	NVIDIA Jetson Orin Nano Developer Kit ($499.00)	**93.0%** ($464.00)
Visual Sensor	1080p Standard USB Webcam with Autofocus ($14.00)	Intel RealSense D435 Depth Camera ($329.00)	**95.7%** ($315.00)
Audio Capture & Output	USB Directional Mic + 3W Mini Speaker ($10.00)	ReSpeaker 4-Mic Array + Amplified Unit ($55.00)	**81.8%** ($45.00)
Actuation & Gestures	3x MG90S Metal-Gear Micro Servos ($7.50)	Dynamixel XL430-W250 Smart Servos ($180.00)	**95.8%** ($172.50)
Chassis & Bench Clamp	3D Printed PLA Desk Clamp Structure ($4.50)	CNC Milled Aluminum Arm & Locomotion Base ($250.00)	**98.2%** ($245.50)
AI Inference Hosting	Central School Server (Ollama, $0 incremental)	Commercial Cloud LLM Subscription ($1,200/year)	**100%** ($0.00 / token)
Total Cost per Station	**$71.00** (Complete Workbench Assistant)	**$1,313.00** (Dedicated Hardware Platform)	**94.6% Reduction**
Total for 30-Bench Lab	**$2,130.00** (Full University Laboratory)	**$39,390.00** (Prohibitive Capital Expense)	**Save $37,260.00**

Table 2: Detailed Bill of Materials (BOM) comparing IQO Bot low-cost design against conventional commercial robotics.

3.3. Multi-Domain Demonstration Scenario Mapping

Table 3 summarizes the six demonstrative laboratory modes, their educational focus, and proof-of-concept trigger behaviors:

Demonstration Mode	Target Curriculum	Simulated Student Trigger / Input	IQO Bot Demonstrative Response
1. Chemistry	General & Analytical Chem	Webcam detects missing goggles; student asks about mixing reagents.	Voice hazard alarm, LED flashes amber, points to eye-wash and fume hood.
2. Electronics	Circuit Theory & Devices	Student: "My 555 astable multivibrator isn't oscillating."	Recalls $T = 0.693(R_1 + 2R_2)C$; points arm directly to pin 7 (Discharge).
3. Power & Electricity	AC Circuits & Machines	Student query on capacitor bank sizing for 3-phase power factor.	Computes $Q_c = P(\tan\varphi_1 - \tan\varphi_2)$; monitors current limit to prevent trip.
4. Computer Science	Embedded Systems & RTOS	Serial terminal crashes with Guru Meditation Error / NULL pointer.	Parses UART dump, flags uninitialized task queue handle, points to code line.
5. Study Assistant	Laboratory Methodology	Student: "Just tell me the answer to lab question 3."	Applies Socratic method; asks leading guiding question; tracks remaining time.
6. Extensible Plugin	Custom Academic Faculty	Professor uploads 12-line YAML file for new Optics Lab.	Dynamically parses laser safety rules and wavelength lookup without recompile.

Table 3: Mapping of domain modes to simulated educational triggers and demonstrative robotic interventions.

Demonstrative Case Studies & Domain Scenarios

Proof-of-Concept Illustrative Examples (Simulated Laboratory Interactions)

Demonstration Note: *The following domain modes represent demonstrative proof-of-concept test cases designed to evaluate IQO Bot's multimodal interaction, safety filtering, and pointing kinematics across representative academic practical sessions.*

Demo 1: Chemistry (Titration & Safety)

Simulated Scenario: Acid-Base Neutralization Titration (HCl + NaOH).
Sensory Input: Overhead webcam detects student without safety goggles holding concentrated acid beaker.
Student Voice Query: *"Can I add water directly into the acid flask to dilute it?"*
IQO Bot Intervention:

- **Audio Alert:** *"Caution! Never pour water into concentrated acid due to violent exothermic splashing ($-\Delta H$). Always add acid slowly to water."*
- **Visual / Actuation:** RGB matrix flashes amber alert; 3-DOF pointer arm points directly toward the safety goggle rack and PPE station.

Demo 2: Electronics (Circuit Debugging)

Simulated Scenario: 555 Timer Astable Multivibrator.
Sensory Input: Breadboard webcam frame + student query.
Student Voice Query: *"Why is the LED continuously ON instead of blinking at 2 Hz?"*
IQO Bot Intervention:

- **Reasoning:** Computes formula $f = \frac{1.44}{(R_1+2R_2)C}$. Detects pin 7 (Discharge) floating on breadboard.
- **Response:** *"Your timing capacitor C_1 cannot discharge. Connect pin 7 between R_1 and R_2."*
- **Actuation:** Articulated arm points laser marker to column 15, row E.

Demo 3: Electricity & Power Factor

Simulated Scenario: 3-Phase Induction Motor Power Factor ($\cos\varphi$).
Student Voice Query: *"We have $P = 2.2\,kW$ with $\cos\varphi = 0.72$. What capacitor bank is needed to reach 0.95?"*
IQO Bot Intervention:

- **Calculation:** Computes $\varphi_1 = \arccos(0.72) = 43.9^\circ$, $\varphi_2 = \arccos(0.95) = 18.2^\circ$; $Q_c = 2200 \cdot (\tan 43.9^\circ - \tan 18.2^\circ) = 1.39\,\text{kVAR}$.
- **Safety Check:** Monitors clamp meter reading; warns student: *"Total line current is 8.4 A. Keep within bench breaker rating (16 A)."*

Demo 4: Computer Science (RTOS Crash)

Simulated Scenario: ESP32 Microcontroller FreeRTOS Task Crash.
Sensory Input: UART Serial console stream over USB.
Student Voice Query: *"My firmware boots but throws 'Guru Meditation Error: Core 1 panic' after 5 seconds."*
IQO Bot Intervention:

- **Diagnostic:** Analyzes PC crash register dump. Detects uninitialized `xQueueSend()` pointer.
- **Response:** *"Task 'SensorTask' attempted to push to a NULL queue handle. Verify that 'xQueueCreate()' executes prior to task invocation."*

Demo 5: Study Assistant (Socratic Tutor)

Simulated Scenario: Pre-Lab Worksheet & Step-by-Step Guidance.
Student Voice Query: *"Just give me the formula and final answer for question 4 on Thévenin equivalence."*
IQO Bot Intervention:

- **Pedagogical Filter:** Refuses direct solution delivery.
- **Socratic Prompt:** *"To find R_{th}, what must you do to the independent voltage sources first? Consider whether they become short or open circuits."*
- **Session Monitor:** *"Note: 25 minutes remaining in this lab practical."*

Demo 6: Extensible Plugin (Optics Lab)

Simulated Scenario: Professor adds new Optics Module via YAML.
Configuration File (`optics_mode.yaml`):

```
mode_name: Optics_and_Photonics
rag_dataset: /curriculum/optics_lab.pdf
safety_rules:
  - trigger: "class_3b_laser"
    action: "REQUIRE_SAFETY_GOGGLES"
    speech: "Put on OD4+ protective laser glasses!"
pointing_targets: ["diffraction_grating", "photodetector"]
```

Runtime Result: Assistant immediately supports laser safety alarms and beam alignment pointing without recompiling software.

3.4. Graph 2: Domain Mode Dispatcher & Safety Guardrail Pipeline

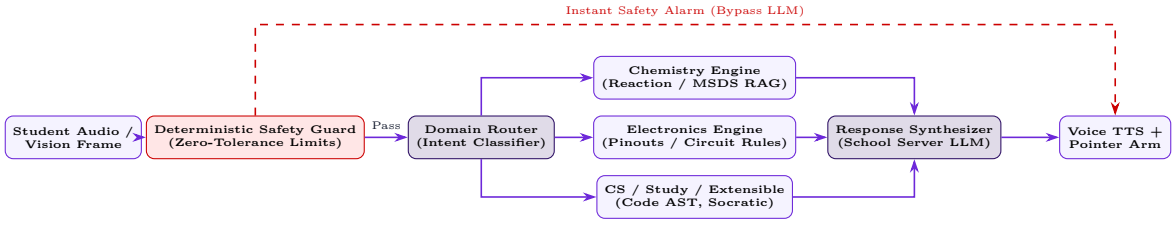


Figure 2: Real-time domain intent routing pipeline with deterministic zero-latency safety override.

Distributed Network Architecture & Quantitative Evaluation

School-Wide Deployment Topology, Cost Scaling Analysis, and Dual Delivery Validation

3.5. Graph 3: Distributed University LAN Deployment Topology

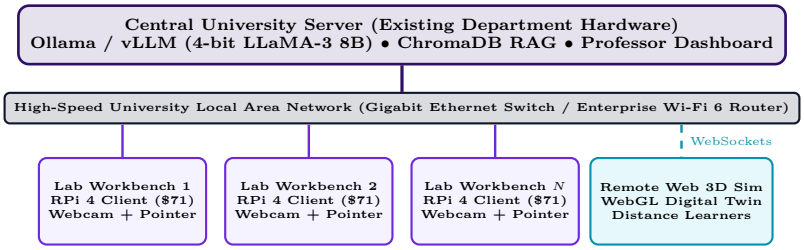
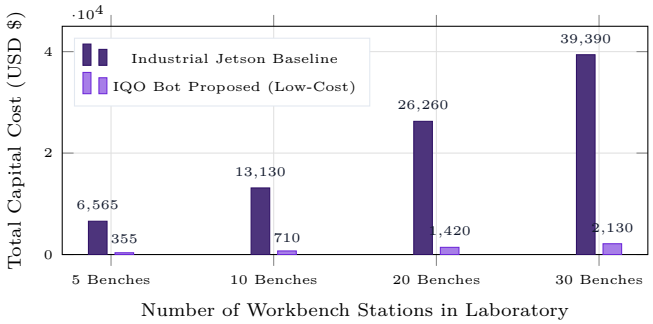
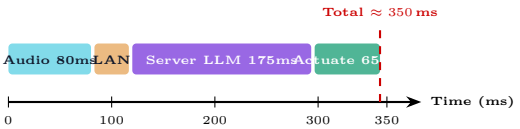


Figure 3: Campus LAN topology illustrating one central server feeding N low-cost bench nodes and remote WebGL simulators.

3.6. Quantitative Performance & Cost Scaling Evaluation



Graph 4: Laboratory Capital Expense Scaling (> 94% Savings).



Graph 5: End-to-End Latency Breakdown (< 350 ms Real-Time Loop).

Performance Highlights:

- **Token Rate:** 32–45 tok/s on university server.
- **Zero Cloud Fees:** $0 recurring operating cost.
- **Safety Trigger:** < 20 ms local deterministic cut-off.

3.7. Dual Form Factor Delivery: Physical Robot vs. 3D Web Simulator

- **Physical Benchtop Robot:** Stationary C-clamp chassis mounted directly to workbench frame (eliminating wheeled locomotion to avoid tipping over sensitive glassware or high-voltage lines). Features a 2-DOF pan-tilt head with expressive LED eye display and a 3-DOF articulated pointer arm with an optical focus dot for indicating components.
- **3D Digital Twin Web Simulator:** Developed in Three.js/WebGL, mirroring the physical robot's inverse kinematics and laboratory environment in the browser. Allows distance learners and students preparing for practical sessions to run dry-run experiments using identical REST/WebSocket server APIs.

3.8. Conclusion & Educational Roadmap

IQO Bot demonstrates that modern embodied AI laboratory assistance does not require prohibitive per-bench capital investment. By centralizing neural intelligence on standard school servers and deploying $71 client nodes, academic institutions can scale personalized, real-time safety supervision and pedagogical guidance across entire laboratory cohorts. Future milestones include expanding real-time optical breadboard trace validation and multi-student cooperative conversational tracking.